WOOJOO NA

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EDUCATION

Northeastern University *Sep. 2024 – Present*
Ph.D. in Electrical and Computer Engineering | Advisor: Jennifer Dy
· *Research Topic: Mathematical foundations of machine learning, optimal transport, and geometrically structured learning algorithms*

Tufts University *Sep. 2021 – Dec. 2023*
· *Graduate Study in Computer Science (Ph.D. Program); M.S. in Computer Science awarded Dec. 2023*
· *Master's Thesis Advisor: Abiy Tasissa*
· *Master's Thesis: Novel Ensemble Learning Method with Applications in Weakly Supervised Learning*

University of Oxford *Oct. 2016 – July 2019*
B.A. (Hons) in Mathematics

RESEARCH INTERESTS

Mathematical foundations of machine learning; optimal transport; geometry and stability of learning updates; optimization; weak supervision; learning theory.

PUBLICATIONS & PREPRINTS

Woojoo Na, Jennifer Dy. "ITSPACE: Monotone Gaussian Optimal Transport Updates."
International Conference on Machine Learning (ICML), 2026. Accepted as a regular paper. OpenReview
- First author and sole student author; led problem formulation, theoretical development, algorithm design, experiments, and writing.

Woojoo Na, Abiy Tasissa. "*RACH-Space*: Reconstructing Adaptive Convex Hull Space with Applications in Weak Supervision."
arXiv preprint, 2023. arXiv
- First author and sole student author; introduced an adaptive convex-hull reconstruction method for weak supervision.
- Reported state-of-the-art performance on the weak-supervision benchmark tasks considered at the time.

RESEARCH EXPERIENCE

Northeastern University *Sep. 2024 – Present*
Graduate Researcher, Advisor: Professor Jennifer Dy
- Developed *ITSPACE*, a monotone Gaussian optimal-transport framework for geometrically structured learning updates, accepted as an ICML 2026 regular paper.
- Led the project end-to-end, including problem formulation, theoretical analysis, algorithm design, experiment implementation, and manuscript writing.
- Current research focuses on optimal-transport-based learning updates, stability and monotonicity of learning dynamics, and scalable mathematically principled ML algorithms.

Data Intensive Studies Center, Tufts University *Sep. 2021 – Dec. 2023*
Graduate Researcher, Advisor: Professor Abiy Tasissa
- Developed *RACH-Space*, an adaptive convex-hull reconstruction algorithm for weak supervision under partial and noisy label information.
- Built the theoretical motivation, implemented the method, and evaluated performance against weak-supervision baselines; reported state-of-the-art results on the benchmark tasks considered at the time.
- Produced a master's thesis and arXiv preprint on weakly supervised learning and ensemble methods.

University of Oxford, Mathematical Institute *July 2019 – Dec. 2019*
Undergraduate Researcher, Advisor: Professor Christophe Petit
- Developed a group-theoretic trapdoor attack against Cayley hash function parameters proposed at the NutMiC 2019 conference.
- Presented the work at the 17^{th} IMA International Conference on Cryptography and Coding. Conference link

University of Oxford, Department of Computer Science *July 2019 – Dec. 2019*
Undergraduate Researcher, Advisor: Professor Thomas Lukasiewicz
- Studied similarities between transformer encoder word embeddings and conceptual-space geometry.
- Visualized embedding structure to investigate low-dimensional features shared by words in the same semantic category.

University of Oxford, Mathematical Institute *June 2018 – Oct. 2018*
Undergraduate Researcher, Advisor: Dr. Andrey Kormilitzin
- Competed in the n2c2 challenge on adverse drug events and medication extraction from electronic health records.
- Implemented CNN and character-level word-embedding models for concept extraction and relation extraction from clinical text.

University of Oxford, Department of Computer Science *June 2018 – Oct. 2018*
Undergraduate Researcher, Advisor: Professor Thomas Lukasiewicz
- Worked on knowledge graph completion for ontology reasoning.

SELECTED TALKS & PRESENTATIONS

"ITSPACE: Monotone Gaussian Optimal Transport Updates" *July 2026*
Accepted paper presentation, International Conference on Machine Learning (ICML 2026), Seoul, Korea

"Novel Ensemble Learning Method with Applications in Weakly Supervised Learning" *Dec. 2023*
Master's Thesis Defense, Tufts University

"Bounding Worst-Case Errors of Weak Supervision Algorithms" *April 2022*
Tufts University Mathematics Graduate Seminar

"Security of Cayley Hash Functions" *Jan. 2020*
Invited Speaker, Ewha Womans University Cryptography and Mathematics Seminar

"Trapdoor Attack on Cayley Hash Function Parameters Proposed at NutMiC 2019 Conference" *Dec. 2019*
17^{th} *IMA International Conference on Cryptography and Coding* Conference link

"A Deep Learning Approach to Concept Extraction from Medical Texts" *Oct. 2018*
Poster Session, University of Oxford Mathematical Institute

HONORS & AWARDS

- Tufts University Linda M. Abriola Graduate Fellowship ($5,000; awarded to 2 graduate students) *2022 – 2023*
- Tufts University Linda M. Abriola Graduate Fellowship ($5,000; awarded to 2 graduate students) *2021 – 2022*
- Best Prize, Poster Session, Republic of Korea Army Start-up Competition *2021*
- Lady Margaret Hall Academic Development Fund for Undergraduate Research (£500) *2018*

TEACHING EXPERIENCE

Tufts University, Department of Computer Science *Jan. 2023 – May 2023*
Teaching Assistant, CS170 Computation Theory

ADDITIONAL EXPERIENCE

Republic of Korea Army, 2nd Artillery Brigade *Feb. 2020 – Aug. 2021*
Sergeant, Fire Direction Operator
- Completed mandatory military service; received three brigade-level awards for excellence and was selected for combined command-post training with the U.S. Army and Republic of Korea Army.

Utpal Shanghvi Global School, Mumbai, India *July 2017 – Aug. 2017*
Project Intern
- Worked on economic sustainability and public-health projects for villagers in Wada, Maharashtra, India.

LEADERSHIP & SERVICE

Oxford Invariants (Oxford Mathematics Society) *April 2018 – April 2019*
Treasurer

Oxford Mathematical Institute, Problem Solving Matters Program *June 2018 – Oct. 2018*
Mentor for Underprivileged High School Students

St. Nicholas Primary School, Oxford, UK *Oct. 2016 – April 2018*
Volunteer Mathematics Teacher for Immigrant Primary Students

TECHNICAL SKILLS & MATHEMATICAL BACKGROUND

Programming	Python, C++, R, JavaScript, MATLAB
ML / Scientific Computing	PyTorch, TensorFlow, NumPy, SciPy, scikit-learn, Git, LaTeX
Mathematics	Optimal transport, optimization, probability, linear algebra, topology, group theory, algebraic geometry